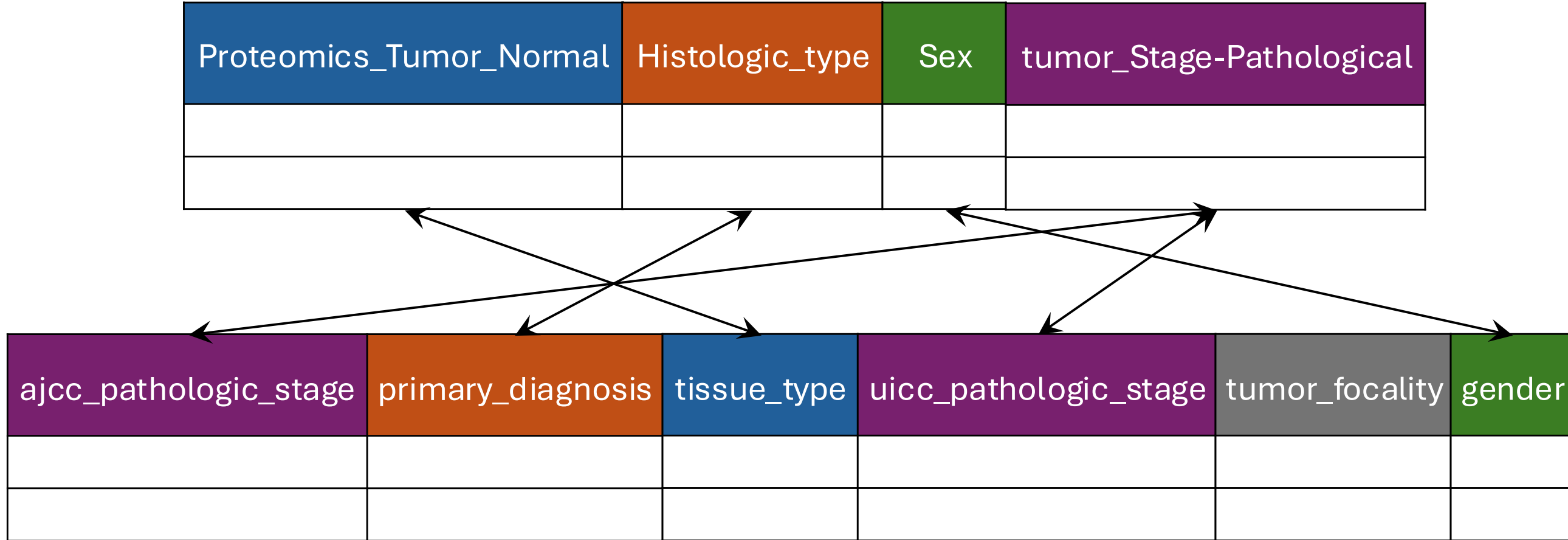


Biomedical Data Integration

- Li et al. [2] carried out a proteogenomic analysis of cancer tumors
- Collected data from 10 studies spanning cohorts and cancer types
- Map each dataset to the Genomic Data Commons (GDC) standard [3]



Schema Matching Identify semantically equivalent columns across distinct data schemas, enabling integration of heterogeneous datasets

Challenge Nuanced semantics, diverse representations of column names and values, requiring biomedical domain expertise, and scaling to schemas with 700+ attributes

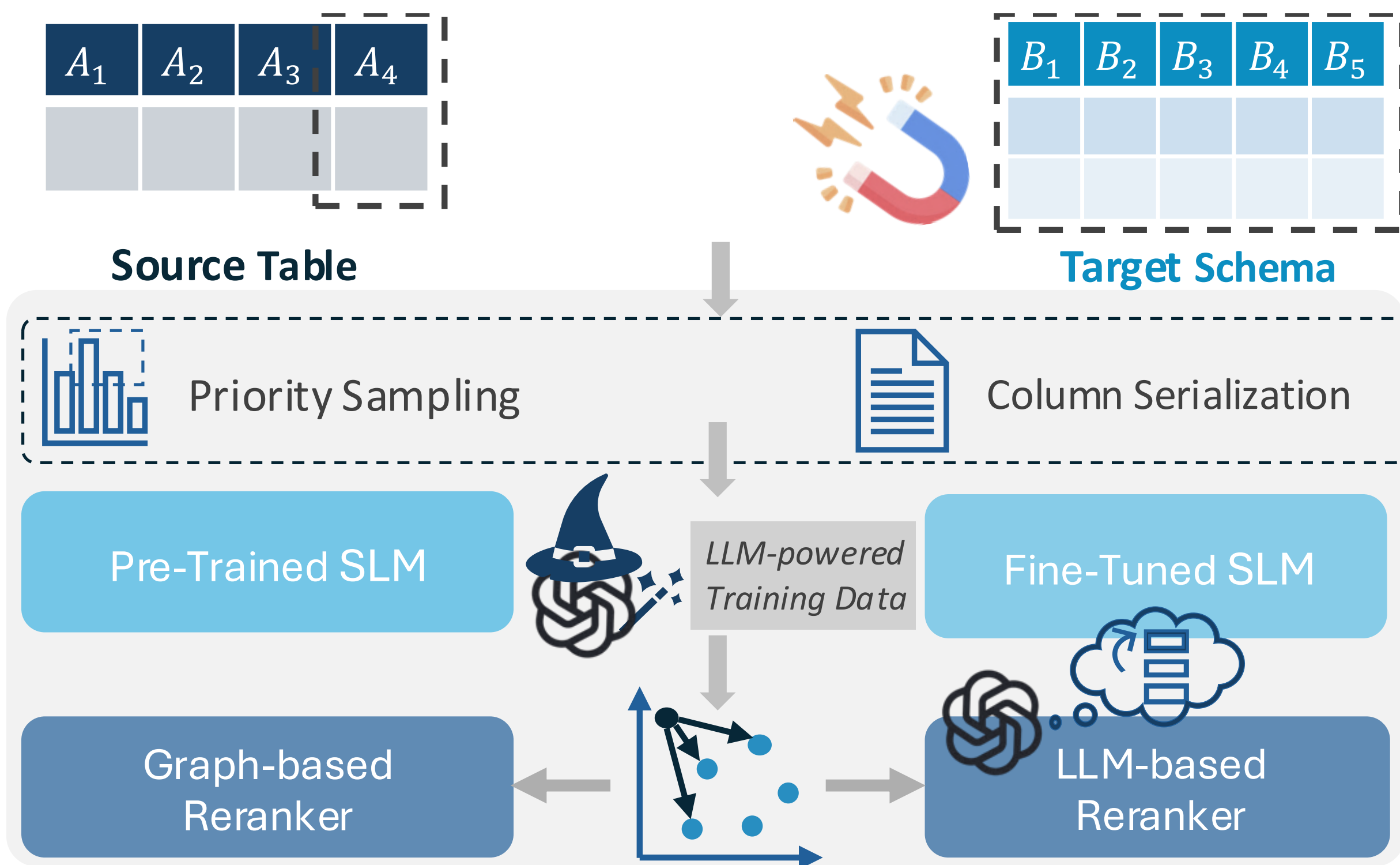
Why Prior Schema Matchers Fall Short?

- Traditional approaches fail in complex scenarios that require semantic understanding
- SLMs embed columns for similarity, but struggle with capturing domain semantics
- Fine-tuned SLMs perform better but need curated training data
- LLMs offer broad knowledge but are limited by context size and cost

Magneto: A Retrieve-Rerank Framework

Retrieval Use SLMs to efficiently retrieve top- k candidates for each source column

Reranking Rerank retrieved candidates using specific criteria



Challenge Represent tabular data in a form that SLMs can process to obtain high-quality column embeddings

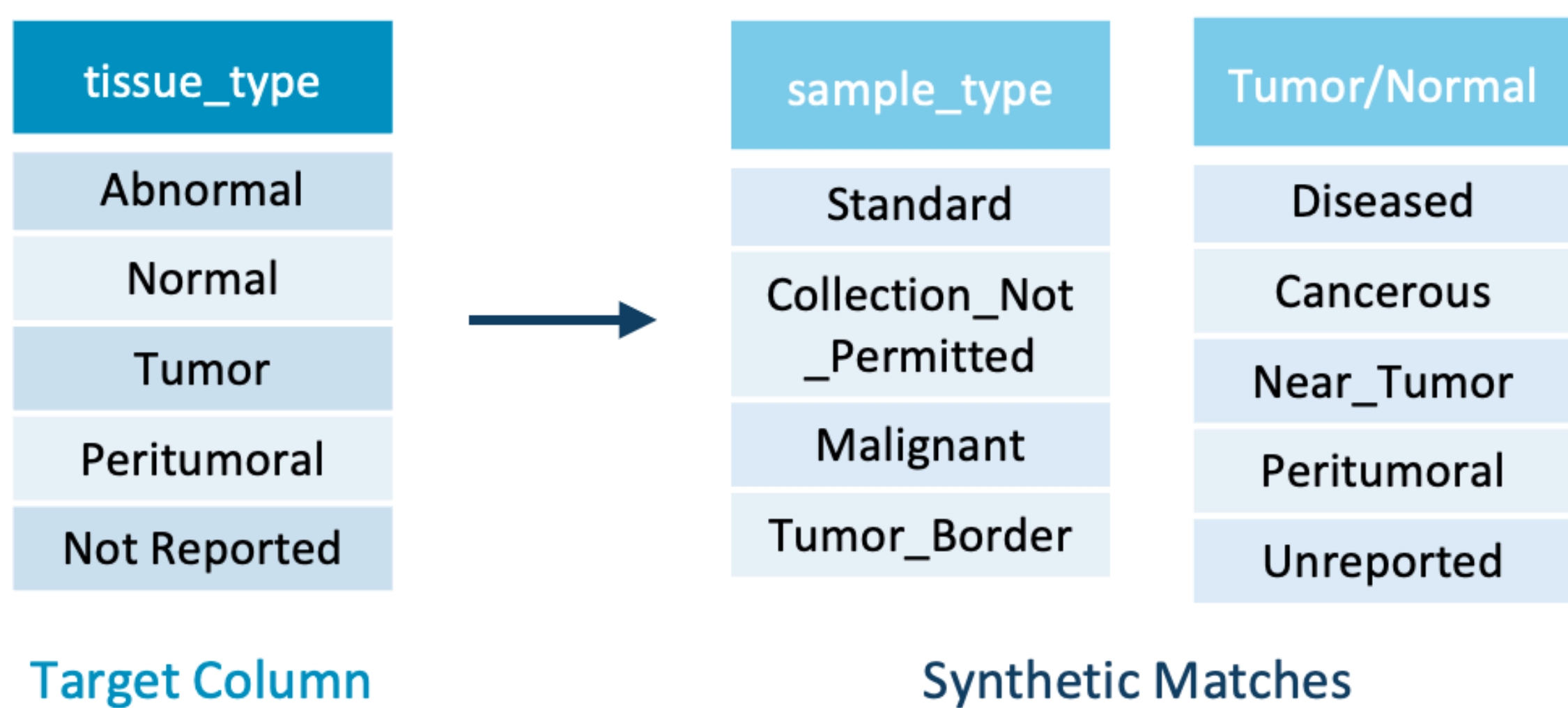
Value Sampling Use priority sampling to balance frequent values with hash-based consistency to increase the likelihood of identifying shared values across columns [1]

Column Serialization Convert table columns into natural language sequences for SLM processing by introducing novel serialization strategies tailored to different contexts

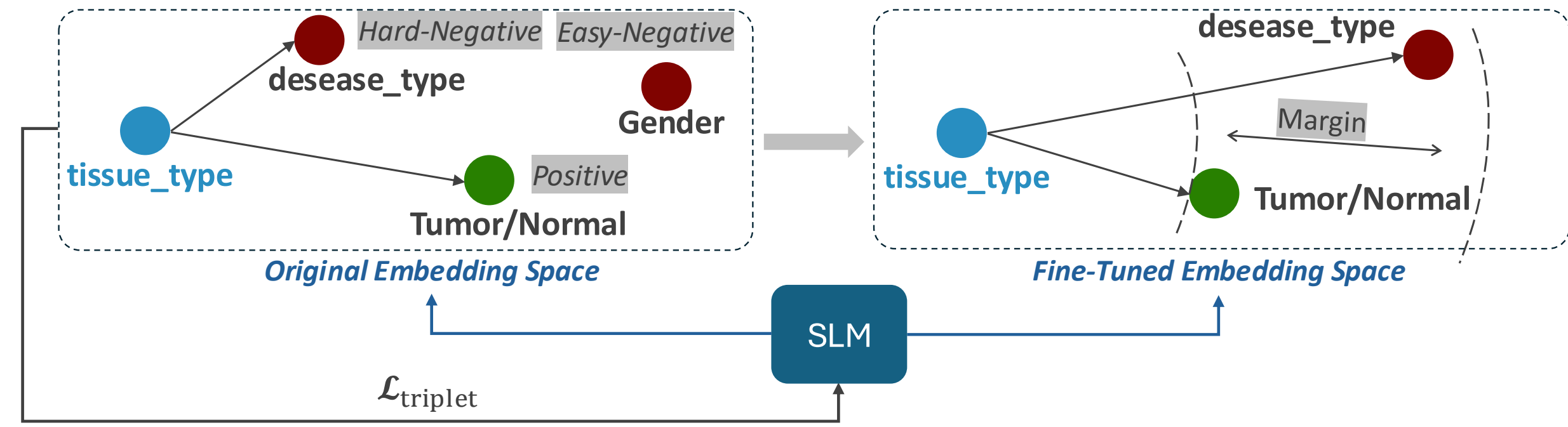
K-Nearest Neighbor Search Use a pretrained SLM to convert columns into embeddings and retrieve top- k candidates

LLM-Powered Fine Tuning

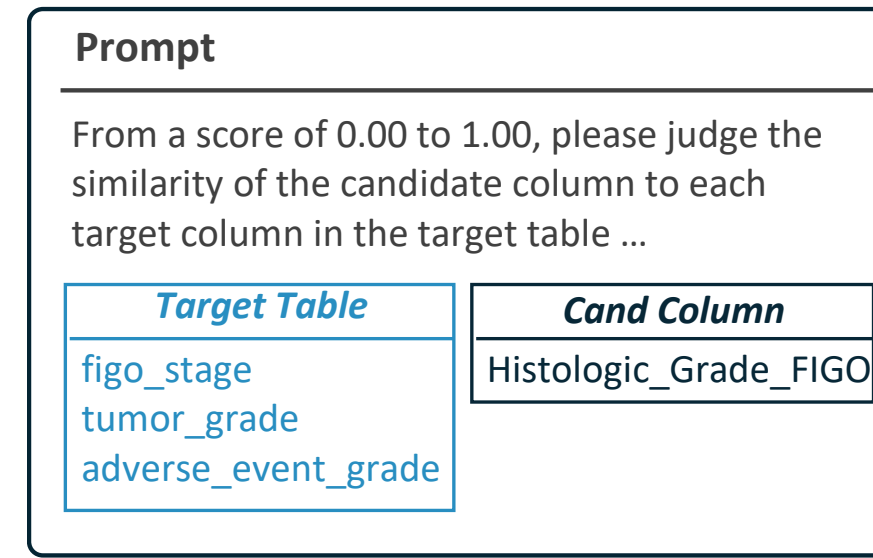
LLM-powered Data Generation Use LLMs to automatically generate semantically coherent synthetic training data needed to fine-tune the small language model



Self-supervised Fine-Tuning Apply contrastive learning using triplet loss and online triplet mining to fine-tune SLMs on these LLM-generated pairs



LLM-Based Reranker



- Given a source column and top- k candidates, LLM reranks target columns by generating pairwise matching scores
- One-shot in-context learning ensures consistent scoring

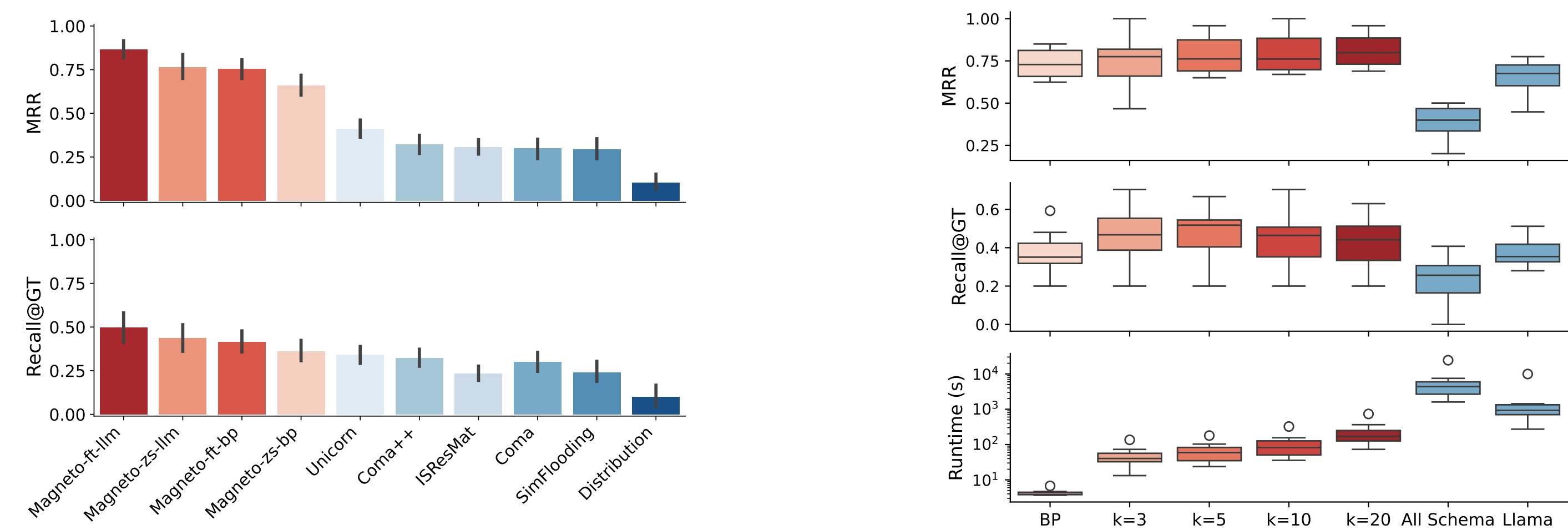
GDC Benchmark

- Based on the Pan Cancer Analysis use case [2]
- Consists of source datasets produced by ten studies (16–213 columns each)
- Mapped to GDC schema: 736 diverse variables
- Gold data created by biomedical researchers from NYU Medical School

Complement existing benchmarks Captures nuanced semantics, scales to large schemas, and reflects real-world applicability

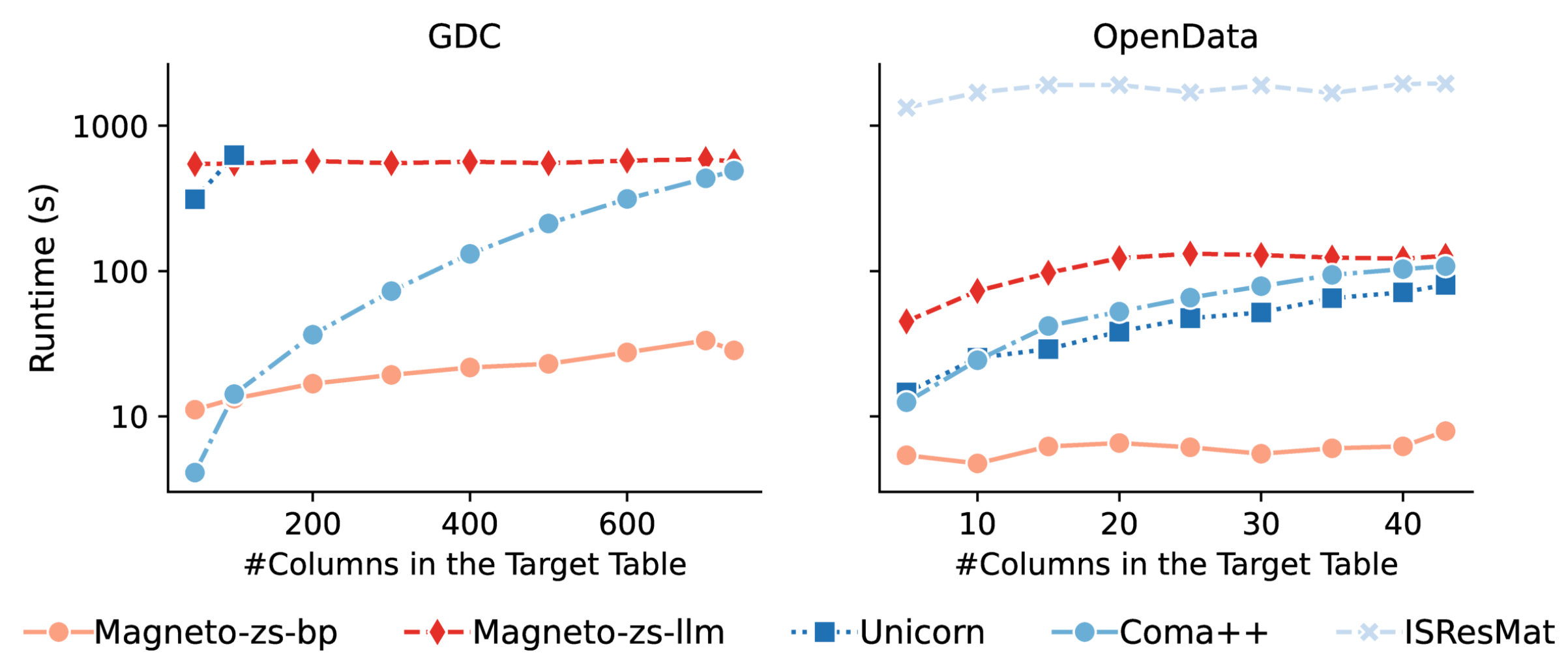
Experimental Results

Magneto attains higher accuracy on the challenging GDC benchmark than traditional, model-based, and supervised approaches



LLM-only approaches are costly and underperform in matching accuracy

Magneto offers alternatives to balance accuracy and efficiency based on user needs



Magneto-zs-bp is significantly faster than the other baselines

Magneto-zs-llm maintains a consistent runtime

Check out the paper for more detailed ablations! Key takeaways are:

- Column sampling and serialization strategies are important
- Magneto** is robust across backbone choices

Acknowledgements

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References

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